

Let's learn about **satellites!**

Space mission and **satellite** design course

Mission management development

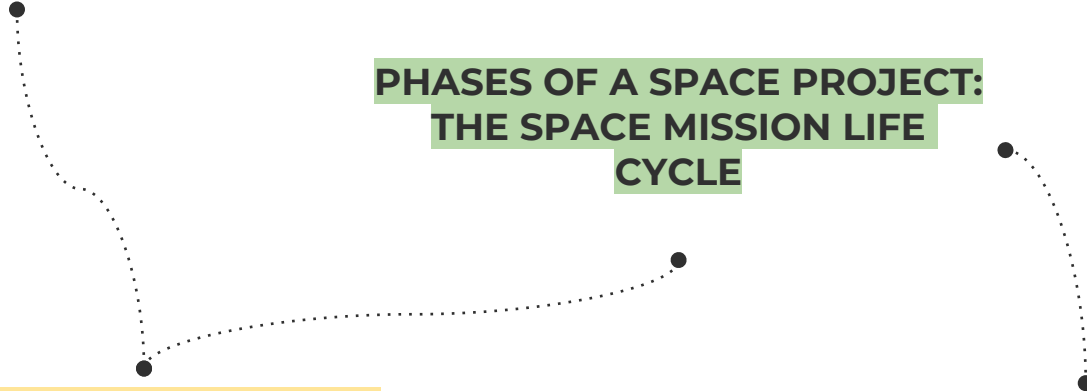
What are we going to talk about today?

MISSION MANAGEMENT

PHASES OF A SPACE PROJECT:
THE SPACE MISSION LIFE
CYCLE

SPACE MISSION DESIGN

LET'S DESIGN A
MISSION!



MISSION MANAGEMENT

What is PROJECT MANAGEMENT?

Project management is the process of planning, organizing, and overseeing the overall execution of a project from the beginning to its end.

Effective project management requires:

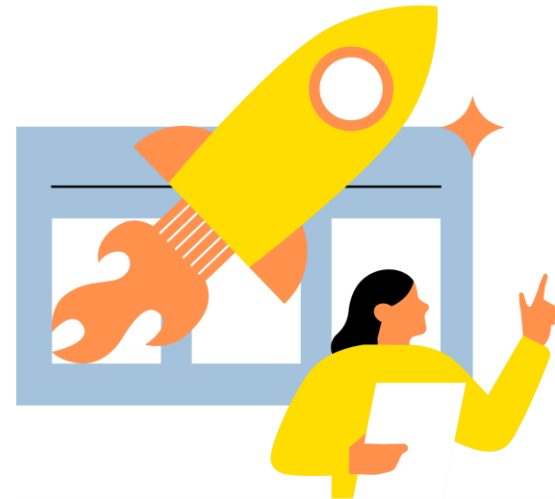
- ★ strong communication skills
- ★ attention to detail
- ★ adapting to ever-changing circumstances.



Project management in the space industry

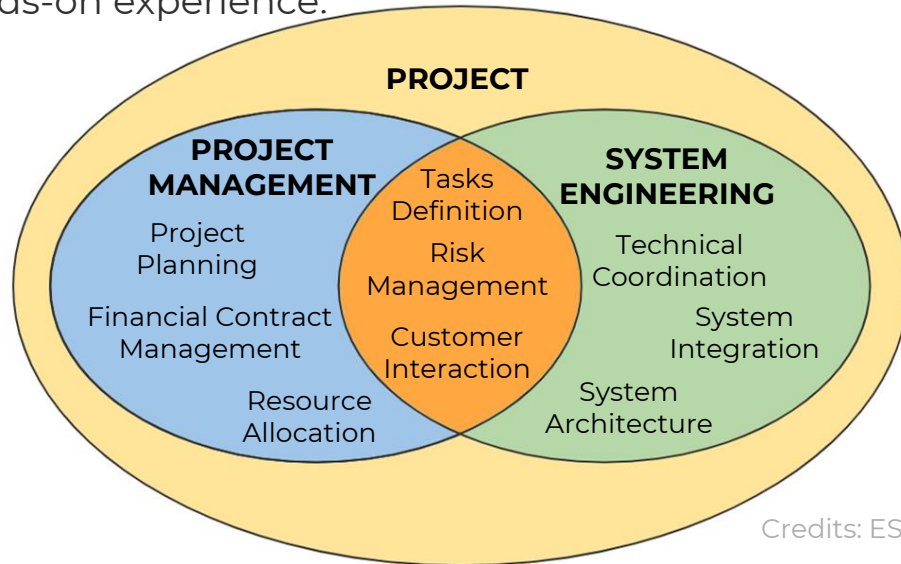
The application of project management concepts and techniques in space-related projects requires to address a variety of issues that are quite unique from those faced on other types of projects.

Project management of the development and execution of space projects involves the **integrated approach** to product definition, development, production, verification, operation, and disposal in response to client's project objectives through various phases of the **space project life cycle**.



Who is the project manager in the space industry?

The space environment is hostile; therefore, reaching and operating in it is particularly difficult. Hence, a good Project Manager is someone who has been a good System Engineer in the past, particularly if they have space engineering hands-on experience.



Credits: ESA

SPACE MISSION DESIGN

What are the goals of our mission and how do we best achieve them?

Which launch vehicle is better to use to get our satellite into orbit?

In which orbit is the satellite going to be deployed?

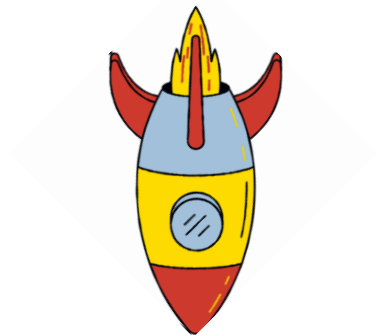


Which payloads are going to be included?

How and when will the “end of life” be performed?

What is space mission design?

Space mission design answer to all this kind of questions!



Usually, we talk about Space Mission Analysis and Design (SMAD) and we refer to it as **“the process of planning and organizing a space mission from its conceptualization to its execution”**.

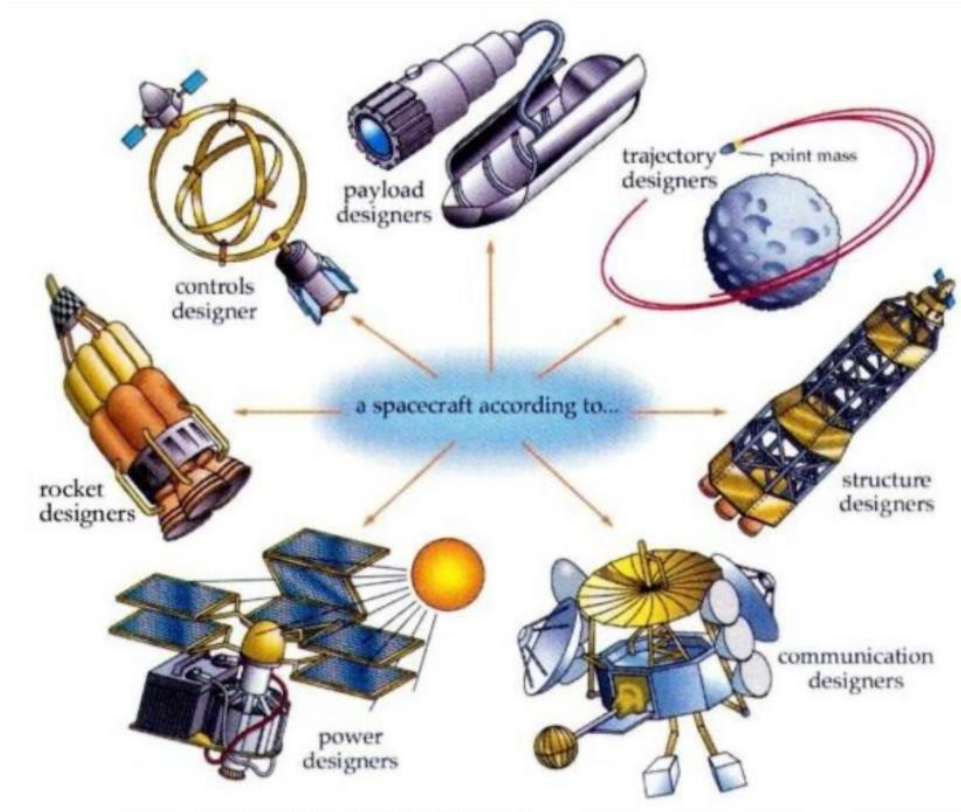
What is space mission design?

Space mission design involves a **team of multidisciplinary experts** that tackle various aspects, including mission objectives, scientific goals, payload selection, trajectory planning, launch vehicle selection, mission duration, communication strategies, and operational considerations.

They **work together to define every aspect of the mission** considering its purpose, scope, and requirements.



Importance of a multidisciplinary team



Steps of SMAD - I

★ Define Objectives

- Define broad objectives and constraints
- Estimate quantitative mission needs and requirements

★ Characterize the Mission

- Define alternative mission concepts
- Define alternative mission architectures
- Identify system drivers for each
- Characterize mission concepts and architectures

Steps of SMAD - II

★ Evaluate the Mission



- Identify critical requirements
- Evaluate mission utility
- Define baseline mission concept

★ Define Requirements



- Define system requirements
- Allocate requirements to system elements

Steps of SMAD - III

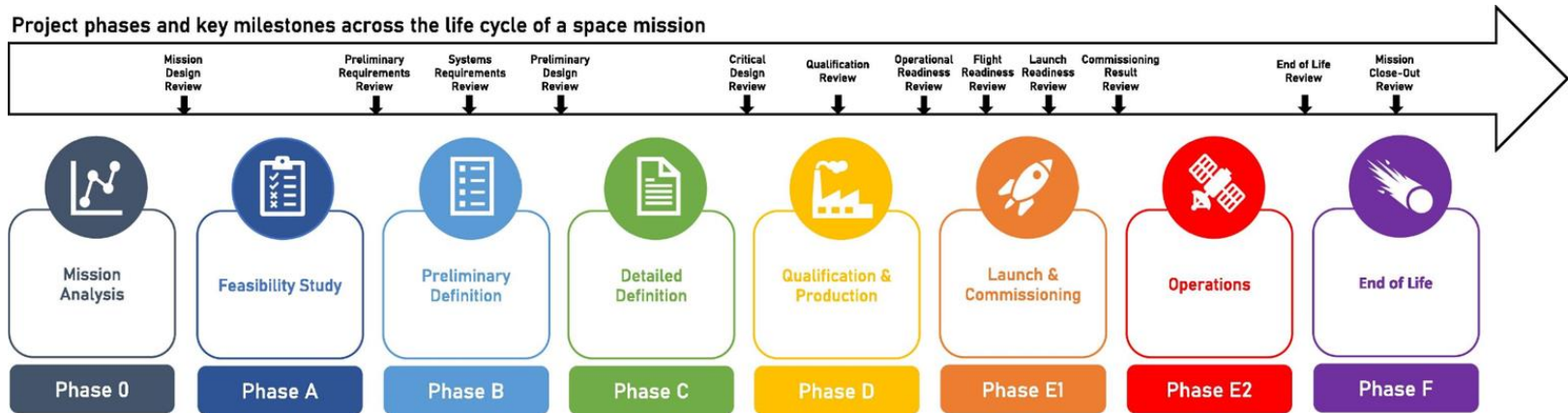
The process described is **ITERATIVE!**

This means that we must repeat it many times for each mission, gradually refining both the requirements and methods of achieving them.



The first several iterations may take only a day, but more detailed requirements will take far longer.

Definition of phases and milestones



Credits: University of Strathclyde

Milestones



MRR



PRR



SRR

Milestones mark points in a project schedule. They can note the start and finish of a project, mark the completion of a major phase of work or anything that's worth highlighting in a project. In a space mission they often coincide with reviews.



PDR



FRR



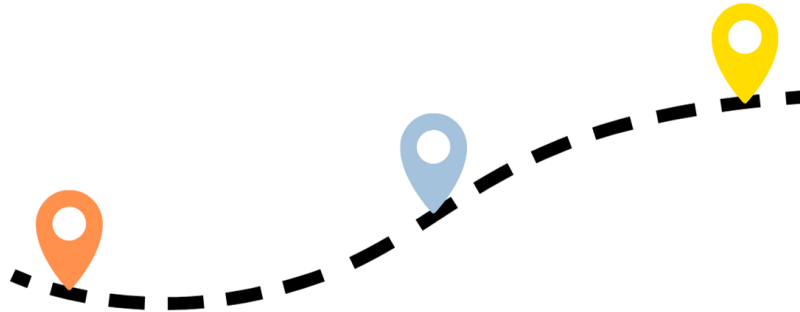
TRR



CDR

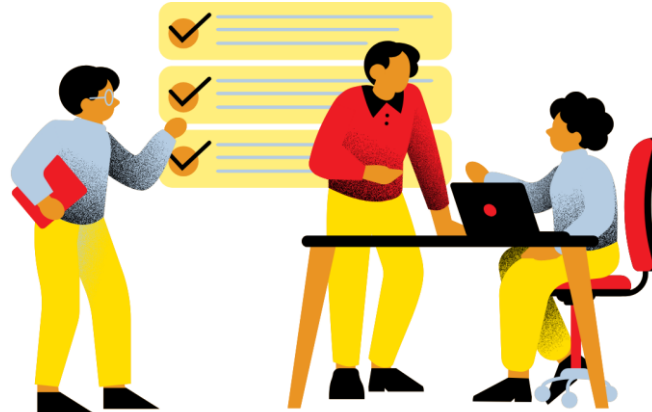
Reviews - I

A review is a thorough overall examination of the technical status of the project that is performed at crucial steps of the programme. Reviews assess the work performed by all participants in a project and they identify potential problems at an early stage, so, they can result in decisions or recommendations to the project management on how to solve these problems. In addition, **the outcome of project reviews can serve to measure the suppliers' progress against prescribed requirements.**



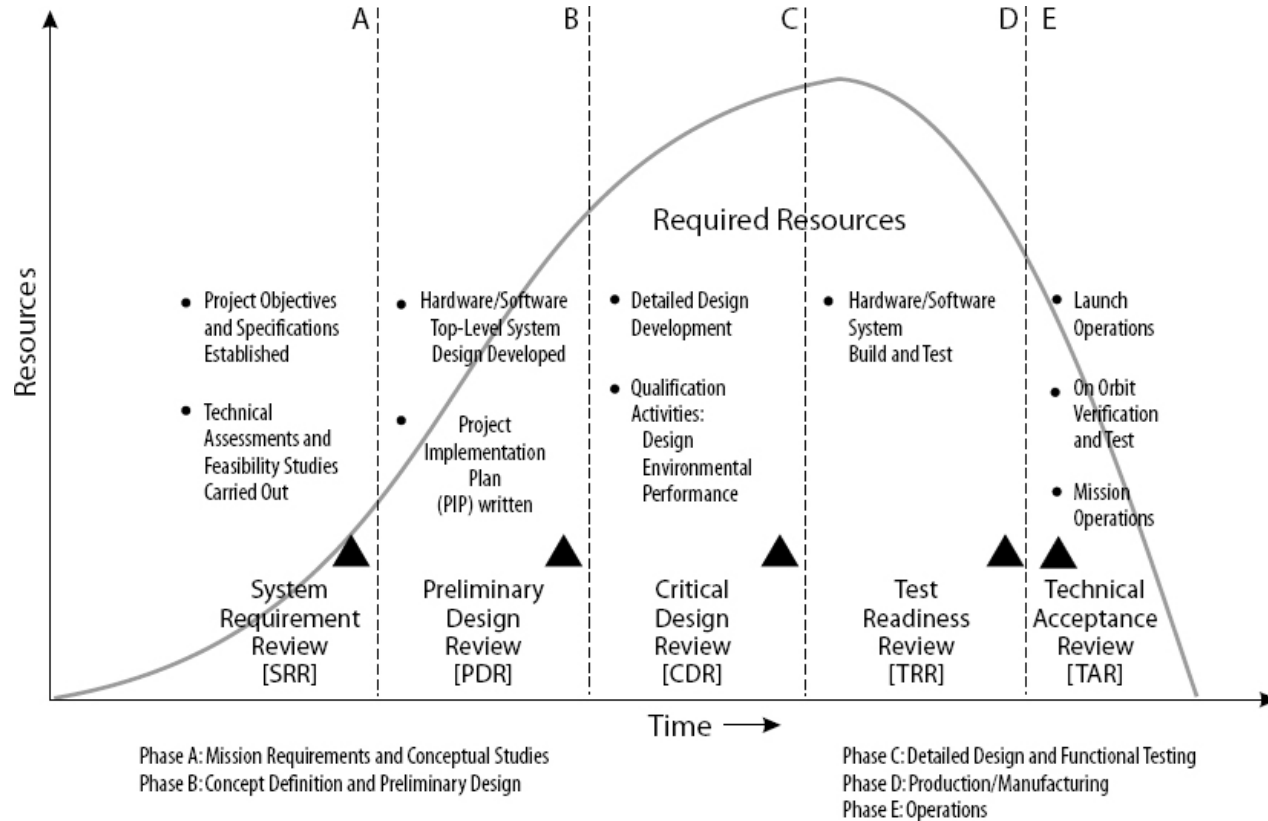
Reviews - II

A project review shall avail itself of strong, **independent expertise** which is **impartial to the interests of both the customer's project team and the supplier**. Review groups shall have access to all the information necessary for them to carry out their tasks.

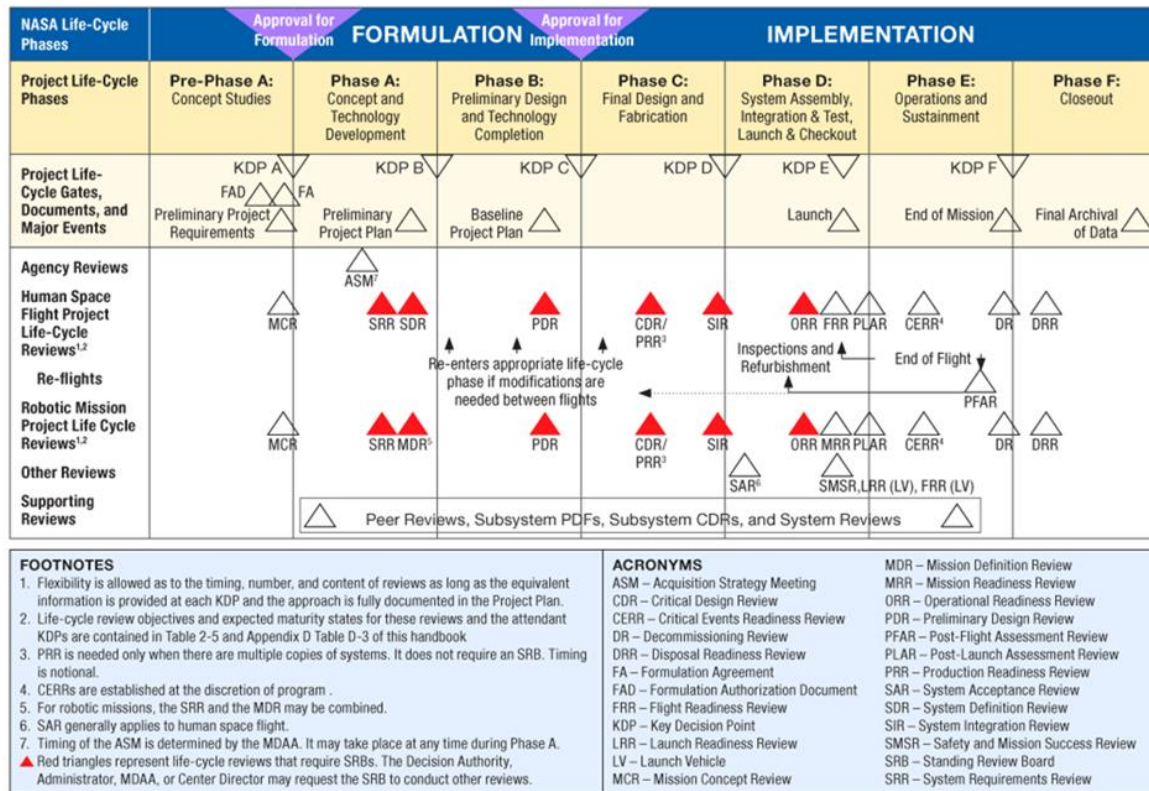


PHASES OF A SPACE PROJECT: THE SPACE MISSION LIFE CYCLE

Space project life cycle



Space project life cycle - NASA



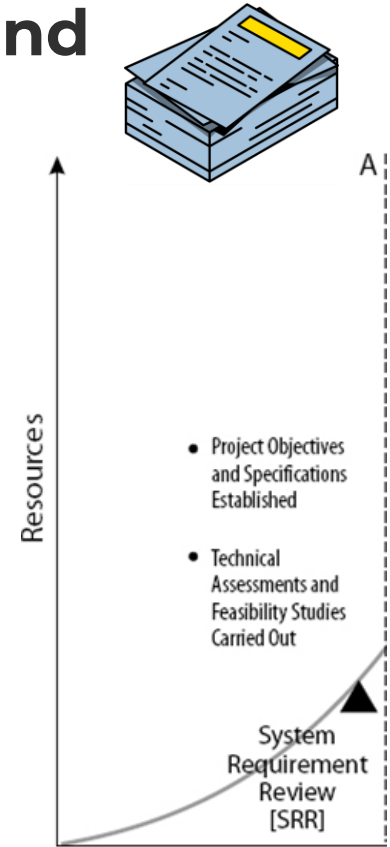
Credits: NASA Space Flight Project Life Cycle from NPR 7120.5E

Phase A: Mission Requirements and Conceptual Studies - I

Purpose: to produce a spectrum of ideas and alternatives for the mission and to develop a proposed mission/system architecture that is credible and responsive to program expectations, requirements, and constraints on the project, including resources.

- ★ First, a team analyses a range of mission concepts that can fall within technical, cost, and schedule constraints.

At this stage the emphasis is on establishing feasibility and desirability rather than optimality.

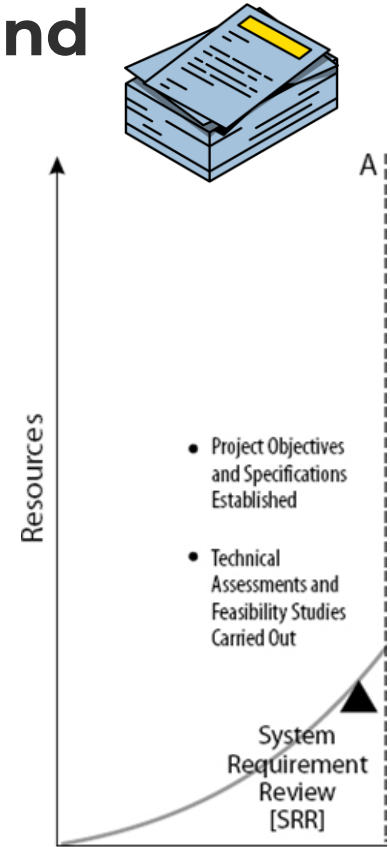


Phase A: Mission Requirements and Conceptual Studies - II

- ★ Second, the team's effort focuses on analyzing mission requirements and establishing a mission architecture. Activities become formal, and the emphasis shifts toward optimizing the concept(s) design.

Systems engineers are heavily involved during this phase in the development and assessment of the architecture and the **allocation of requirements to the architecture elements**.

System functional and performance requirements, along with architectures and designs, become firm as system tradeoffs and subsystem tradeoffs iterate back and forth.



Phase A - Reviews

Stage of activity just completed	Review name	Output for the system	Output for subsystems	Output for equipment
Identification of user requirements and initial concepts	MDR (Mission Definition R.)	Confirmation of the mission requirements	/	/
Initial requirements converted into overall system requirements, i.e. mission feasibility and validation of system architecture	PRR (Preliminary Requirements R.)	Confirmation of system feasibility and functional specification released. Requirements on system interfaces established by the first level customer	Allocation of subsystem functional requirements	/
System technical specification established	SRR (System Requirements R.)	Assessment of preliminary performance based on conformance with system functional requirements. System technical specification released, with inclusion of external interfaces.	Allocation of technical requirements on each subsystem	Identification of technical requirements for critical technologies

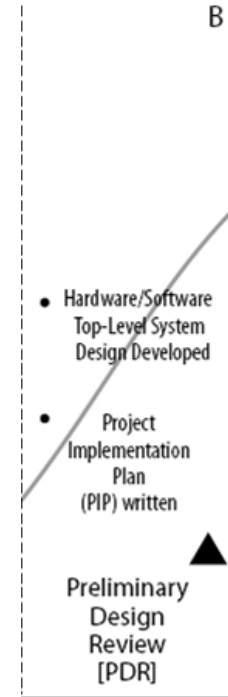
Phase B: Concept Definition and Preliminary Design



Purpose: to complete the technology development, engineering prototyping, heritage hardware and software assessments, risk-mitigation activities and the preliminary design.

Systems engineers are involved in this phase to **ensure** the **preliminary designs of the various systems** will work together, **are compatible**, and are likely to meet the customer expectations and applicable requirements.

Phase B is where the top-level requirements are defined. While the requirements should be baselined in Phase A, changes resulting from the trade studies and analyses in late Phase A and early Phase B may result in changes or refinement to system requirements.



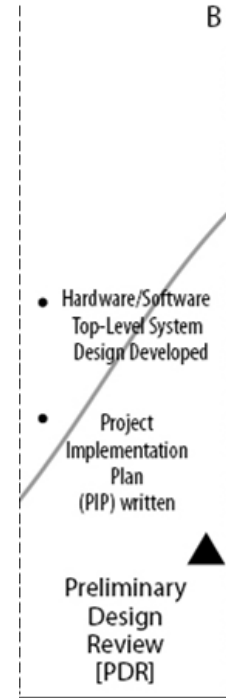
Phase B: Concept Definition and Preliminary Design



In this phase, it is important to **validate design decisions against the original goals** and objectives. All aspects of the life cycle should be considered, including design decisions that affect training, operations resource management, human factors, safety, habitability and environment, and maintainability and supportability.

The Phase B baseline consists of a collection of evolving baselines covering technical and business aspects of the project: subsystems requirements and specifications, designs, schedules, cost projections, etc..

From this point on, almost all changes to the baseline are expected to represent successive refinements, not fundamental changes. **Significant design changes at and beyond Phase B become increasingly expensive.**



Phase B - Reviews

Stage of activity just completed	Review name	Output for the system	Output for subsystems	Output for equipment
Preliminary design established	PDR (Preliminary Design R.)	Assessment of performance based on analysis results, establishment of technological readiness and compatibility between design and customer requirements, approval of project plans (such as qualification plan, verification and test plan) and standards. Compatibility between design and customer requirements, confirmation of verification approach, special models design. Internal interfaces established	Controlled configuration. Test and verification methods defined. Release of technical subsystem specifications	Allocation of technical requirements on each equipment. Test and verification methods defined.

Phase C: Detailed Design and Functional Testing



Purpose: to complete and document the detailed design of the system (and its associated subsystems, including its operations systems) that meets the detailed requirements. Generate final designs for each system structure product.

During this phase, **activities are performed to establish a complete design**. Trade studies continue and results are used to validate the design against project goals and objectives.

Engineering test units more closely resembling actual hardware are built and tested to establish confidence that the design will function in the expected environments.



Phase C: Detailed Design and Functional Testing



Engineering speciality and crosscutting analysis results are integrated into the design.

Systems engineers are involved in this phase to ensure the final detailed designs of the various systems will work together, are compatible, and are likely to meet the customer expectations and applicable requirements.

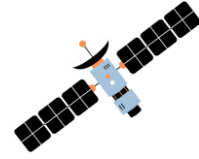
At each step in the successive refinement of the final design, corresponding **integration and verification activities are planned in greater detail**. During this phase, technical parameters, schedules, and budgets are closely tracked to ensure that undesirable trends (such as an unexpected growth in spacecraft mass or increase in its cost) are recognized early enough to take corrective action.



Phase C - Reviews

Stage of activity just completed	Review name	Output for the system	Output for subsystems	Output for equipment
Development testing completed.	CDR (Critical Design R.)	Design justification file: e.g. integration and test of functional model, design analyses and preliminary version of users documentation	"Build-To", "Buy-To" documents and test procedures	"Build-To", "Buy-To" documents and test procedures

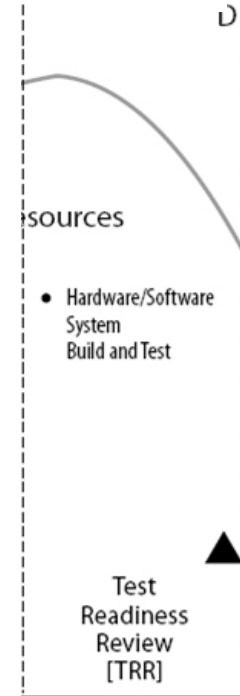
Phase D: Production/Manufacturing



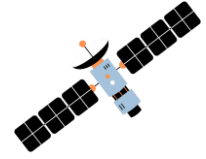
Purpose: to fabricate or produce hardware, code software and to assemble, integrate, verify and validate the system.

Activities include **testing the flight system** to expected environments within margin, **updating operational procedures**, rehearsals and training of operating personnel and crew members, and **implementation of the logistics** and spares planning.

System engineering is involved in all aspects of this phase including answering questions and work any interfacing issues that might arise, providing advice, resolving issues, assessing results of the verification and validation tests, and providing information to decision makers for go/no-go decisions.



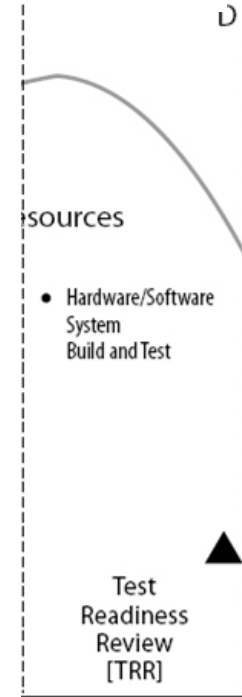
Phase D: Production/Manufacturing



The planning for Phase D activities is generally initiated in Phase A.

The planning for the activities should be performed as early as possible since **changes at this point can become costly.**

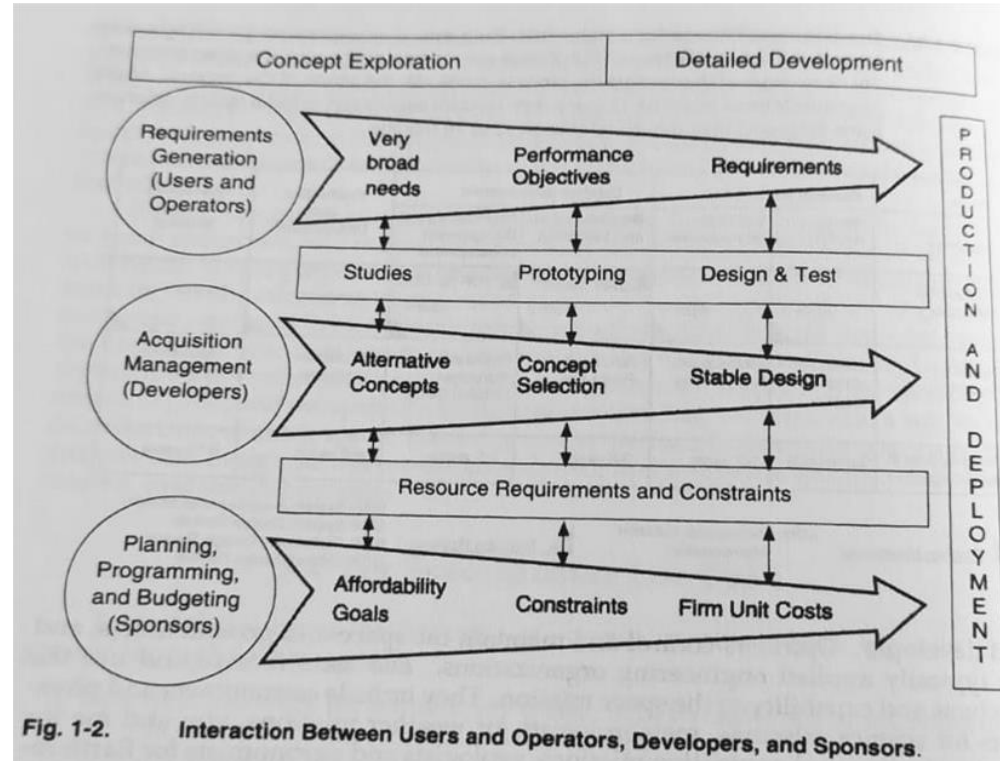
Phase D concludes with a system that has been shown to be capable of accomplishing the purpose for which it was created.



Phase D - Reviews

Stage of activity just completed	Review name	Output for the system	Output for subsystems	Output for equipment
Compatibility and readiness of space element for flight	FRR (Flight Readiness R.)	Confirm proper performance with mission requirements and interfaces with all mission elements	"As-built" configuration confirmed	/

From concepts to production



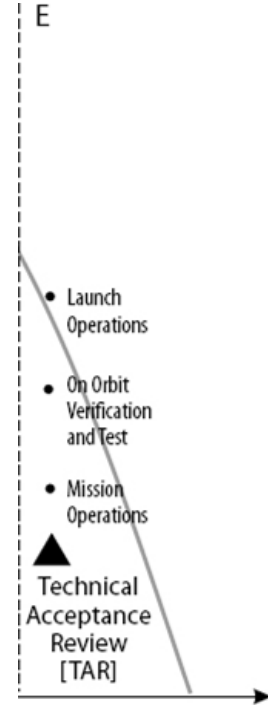
Phase E: Operations



Purpose: to conduct the prime mission to meet the initially identified needs and to maintain support for that needs. The products of this phase are the results of the mission and performance of the system.

Specialty engineering disciplines, like maintainability and logistics servicing, will be performing tasks during this phase.

This phase encompasses the evolution of the system only insofar as that evolution does not involve major changes to the system architecture. Changes of that scope constitute new “needs,” and the project life cycle starts over.

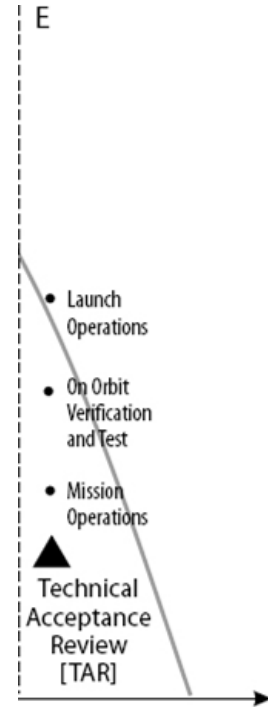


Phase E: Operations



In this phase systems engineers continue to play a role since integration often overlaps with operations for complex systems and systems with complex sustainment needs or human involvement will likely require evaluation and adjustments that may be beyond the scope of operators to perform.

Systems engineering personnel also may be involved in in-flight anomaly resolution. Additionally, software development may continue well into Phase E. For example, software for a planetary probe may be developed and uplinked while in-flight. Another example would be new hardware developed for space station increments.



Phase E - Reviews

Stage of activity just completed	Review name	Output for the system	Output for subsystems	Output for equipment
Test and verification of actual performance versus mission requirements after useful time of flight operation or service	FQR (Flight Qualification R.)	Operational qualification of system or in-orbit check-out completed and capability to fulfil intended mission confirmed	/	/
End of useful time of operation or service	EOLR (End of Life R.)	Condition and procedures for disposal, retrieval, re-entry	Lessons learned	Lessons learned

Mission documents

For each project, a **detailed list of deliverable documents for the important reviews is defined** in the Document Requirements List (**DRL**), which usually **forms part of the contract**.

Documents presented to a review are used to substantiate the design to be reviewed and to amend, for major reviews, after adequate approval, the corresponding frozen baseline.

Documents presented at qualification and acceptance reviews are used to demonstrate completion of relevant verification.



Critical Design Review documents

Examples of documents that can be required for CDR:

- ★ Preliminary drawings and associated lists
- ★ Design analysis reports and development test reports
- ★ Subsystem/system components/prime items integration plans/procedures
- ★ Qualification and test plans and procedures
- ★ Special tooling and support equipment definitions
- ★ Special facility requirements documents



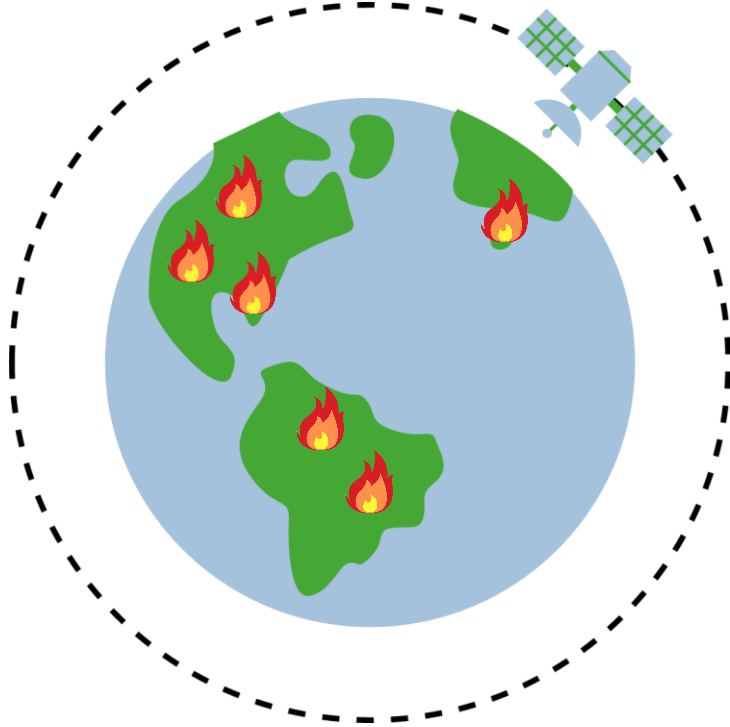
Concurrent Engineering: a new approach to space mission design



LET'S DESIGN A MISSION!

Credits: 1. "Space Mission Analysis and Design", J. R. Wertz and W. Larson
(Ed.), Microcosm Press/Kluwer Academic Publishers (1992)
2. Richard G. Marsden, ESA/SCI-SH, Alpbach Summer School 2002

FireSat



Mission Statement



Because forest fires have an increasing impact on recreation and commerce and ever higher public visibility, the US needs a more effective system to identify and monitor them.

In addition, it would be desirable (but not required) to monitor forest fires for other nations; collect statistical data on fire outbreaks, spread, speed, and duration; and provide other forest management data.

Ultimately, the Forest Service's fire-monitoring office and rangers in the field will use the data.

Data flow and formats must meet the needs of both groups without specialised training and must allow them to respond promptly to changing conditions.

Define objectives - I

Step A: Define broad objective and constraints

Primary Objective:

- ★ To detect, identify, and monitor forest fires throughout the US, in near real time

Secondary Objectives:

- ★ To demonstrate to the public that positive action is underway to contain forest fires
- ★ To collect statistical data on the outbreak and growth of forest fires
- ★ To monitor forest fires for other countries
- ★ To collect other forest management data

Define objectives - II

Step B: Estimate quantitative mission needs and requirements

To transform mission objectives into mission requirements, we need to consider 3 broad areas:

- ★ **Functional Requirements:**

- how well the system has to perform to meet its objectives.

- ★ **Operational Requirements:**

- how the system operates
- how the users interact with the system to achieve the mission's broad objectives

- ★ **Constraints:**

- limitations imposed on system designer by cost, schedule, and implementation techniques

Define objectives - III

Functional Requirements

- ★ Performance
- ★ Coverage
- ★ Responsiveness
- ★ Secondary mission

Constraints

- ★ Cost
- ★ Schedule
- ★ Political
- ★ Interfaces
- ★ Development Constraints

Operational Requirements

- ★ Duration
- ★ Availability
- ★ Survivability
- ★ Data Distribution
- ★ Data Content, Form, and Format

Characterize the mission - I

Step C: Define alternative mission concepts

★ **Data Delivery:**

- how mission and housekeeping data are generated or collected, distributed, and used (How is imagery collected? How are forest fires identified? How are the results transmitted to the fire fighting teams?)

★ **Communications Architecture:**

- how the various components of the system talk to each other

★ **Tasking, Scheduling, and Control:**

- how the system decides what to do in the long term and the short term (e.g. Which sensors are active and when is data being transmitted and processed?)

★ **Mission Timeline:**

- overall schedule for planning, building, deployment, operations, replacement, and end-of-life

Characterize the mission - II

Step E: Identify system drivers for each

Some common system drivers:

★ **Size:**

- Driver limited by e.g. shroud or bay size (shuttle); available weight
- Driver limits e.g. payload size

★ **On-Orbit Weight:**

- Driver limited by e.g. launch vehicle; altitude
- Driver limits e.g. payload weight; survivability; design & manufacturing cost

★ **Power:**

- Driver limited by e.g. size; weight
- Driver limits e.g. payload and bus design; on-orbit lifetime

★ **Data Rate:**

- Driver limited by e.g. storage; processing; antenna size
- Driver limits e.g. information sent to end user; need for onboard processing

Characterize the mission - III

Step F: Characterize mission concepts and architectures

The mission concept characterisation process consists of the following steps:

- ★ Define the preliminary mission concept
- ★ Define the subject characteristics
- ★ Determine the orbit characteristics
- ★ Determine payload size and performance
- ★ Select mission operations approach (comms architecture; operations; ground system)
- ★ Design S/C bus to meet payload, orbit and communications requirements
- ★ Select launch and orbit transfer system
- ★ Determine deployment, logistics, and end-of-life strategies
- ★ Provide costing estimate

- ★ **ITERATE !**

Evaluate the mission - I

Step H: Evaluate mission utility

How well do different mission options satisfy the fundamental mission objectives?

In the case of FireSat, key mission evaluation questions would be:

- ★ Which FireSat requirement dominates the system design or is the most difficult or expensive to meet?
- ★ How well can FireSat detect and monitor forest fires, and at what cost?
- ★ Should the FireSat mission evaluation proceed, and if so, which alternatives should we pursue?

Evaluate the mission - II

Step H: Evaluate mission utility

Mission Utility Analysis quantifies mission performance as a function of design, cost, risk, and schedule.

Typically, two types of quantities are involved:

- ★ **Performance Parameters**

- quantify how well the system works (e.g., coverage statistics, instrument resolution as function of viewing angle)

- ★ **Measures of Effectiveness (MoE)**

- quantify directly how well the system meets the mission objectives (e.g., for FireSat, the MoE is a numerical estimate of how well the system can detect forest fires)

Define requirements

Step J: Define system requirements

- ★ **Performance:**

- 4 temperature levels; 30 m resolution; 500 m location accuracy

- ★ **Coverage:**

- Daily coverage of n million square kilometers within the US

- ★ **Responsiveness:**

- Send registered mission data within 30 min to up to 50 users

- ★ **Availability:**

- 98%, 3-day maximum outage

- ★ **Data Distribution:**

- Up to 500 fire-monitoring offices + 2000 rangers worldwide (max. of 100 simultaneous users)



**THANK YOU FOR YOUR
ATTENTION**

References

- ★ ECSS-M-30-01A - 1 September 1999
- ★ “Space Mission Analysis and Design”, J. R. Wertz and W. Larson (Ed.), Microcosm Press/Kluwer Academic Publishers (1992)
- ★ SEH 3.0 NASA Program/Project Life Cycle